



A single mutation Asp43Arg was increased 2.5-fold the catalytic activity and maintained the stability of cold-adapted *endo*-1,4-beta glucanase (Ef-EG2) from *Eisenia fetida*

Chikako Kuroki^{a,1}, Yu Hirano^{b,c,1}, Mami Nakazawa^a, Tatsuji Sakamoto^a, Taro Tamada^{b,c,*}, Mitsuhiro Ueda^{a,*}

^a Graduate School of Agriculture, Osaka Metropolitan University, Osaka 599-8531, Japan

^b Institute for Quantum Life Science, National Institutes for Quantum Science and Technology, Chiba 263-8555, Japan

^c Department of Quantum Life Science, Graduate School of Science, Chiba University, Chiba 265-8522, Japan

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ABSTRACT

To improve the specific activity at low temperatures of Ef-EG2 and to maintain thermostability, five mutant enzymes (K273R, N372D, Q387E, N402D, D43R) were produced. The two mutant enzymes (K273R, N402D) lost the cellulase activity. The specific activities of mutant N372D, Q387E, and D43R enzymes were 2.5-fold higher than that of WT enzyme over various temperatures. The denaturation temperatures (T_m) of WT and N372D, Q387E, and D43R enzymes were 55.1 °C, 51.5 °C, 51.2 °C, and 55.6 °C. D43R showed almost the same thermostability as the WT enzyme. The three-dimensional structure of D43R was not significantly different from that of the WT enzyme, but the D43R enzyme lost the ability to bind sodium ions. D43R was introduced the additional electrostatic interaction with Asp55.

Introduction

Recently, bioethanol has been paid attention to as petroleum-alternate fuel because environmental problems, such as the increased of emission of greenhouse gases, are getting worse. The cone or cane juice for the production of bioethanol has a competitive relationship with food materials. Plant-derived lignocellulosic wastes, such as pulp and paper, are more cost-effective materials. It is composed of cellulose, hemicellulose, and lignin. For the efficient use of lignocellulosic materials, it is needed to effectively process of saccharification and fermentation. Ordinarily, the lignocellulosic material is pretreated using alkaline and heat to expose the cellulose. The cellulose was hydrolyzed by acid or fungal cellulases (Ingram et al., 2011, Kovacs et al., 2009), and the hydrolyzed products were converted to ethanol by *Saccharomyces cerevisiae*. However, it requires a cooling step for the fermentation at 25–30 °C by *S. cerevisiae*. If the saccharification and fermentation are carried out simultaneously at 25–30 °C, the optimal growth temperature for yeast, the energy required for bioethanol production can be reduced. The simultaneous saccharification and fermentation (SSF) is an eco-

friendly process (Ueda et al., 2018).

The cold-adapted enzymes have great potential value in biotechnological aspects. The enzymes have significant advantages at the level of the specific activity, lower stability, and unusual specificity (Gerday et al., 2000, Siddiqui and Cavicchioli, 2006). These enzymes function at low temperatures, but their stabilities are significantly reduced at high temperatures. Therefore, it is so difficult to use these enzymes for the long term industrially. Recently, we cloned a cold-adapted *endo*-1,4-β-glucanase from the earthworm *Eisenia fetida* (Ef-EG2) (Ueda et al., 2014). It was shown this enzyme could work at 25–30 °C using SSF process and exhibited significant activity at lower temperature (10 °C and 20 °C for 38%, 57% of the activity at optimal 40 °C).

The stability of proteins is dominated by disulfide bonds, hydrophobic core, metal binding, and salt bridges (Jaeqxa, Oomensab, and Rijs, 2013, Pucci and Roman, 2017). Salt bridges are interactions between positive amino acid residues (Arg or Lys) and negative amino acid residues (Glu or Asp), and their strength is shown as the sum of electrical interactions and hydrogen bonds (Gumpena et al., 2012). The activity of the cold-adapted xylanase was improved by removing two salt bridges

* Corresponding authors at: Institute for Quantum Life Science, National Institutes for Quantum Science and Technology, Chiba 263-8555, Japan (T. Tamada), and Graduate School of Agriculture, Osaka Metropolitan University, Osaka 599-8531, Japan (M. Ueda).

E-mail addresses: tamada.taro@qst.go.jp (T. Tamada), m.ueda@omu.ac.jp (M. Ueda).

¹ Chikako Kuroki and Yu Hirano have contributed equally as the first author in this paper.

(Wang et al., 2016). Also, the addition of the new salt bridge improved enzyme thermostability (Ban et al., 2020). Therefore, the salt bridge is crucial as a tool to improve enzyme function. In this study, we tried to adding a salt bridge to the cold-adapted *endo*-1,4- β -glucanase from *E. fetida* and to improve its specific activity and thermostability.

Materials and methods

Chemicals

Site-specific mutagenesis was purchased PrimeSTAR Mutagenesis Basal kit (Takara Bio, Shiga, Japan). Plasmid extraction kit was purchased PureYield Plasmid Midiprep System 25 preps (Promega Co., WI, USA). Carboxymethyl cellulose sodium salt (CMC) was obtained from Nakalai Tesque, Inc. (Kyoto, Japan).

Determination of mutagenesis sites

Based on the crystal structure obtained in the previous study (Arimori et al., 2013), we determined the mutagenesis sites. (1) Introduction of a new salt bridge; if the distance between the side chains of Asn/Gln and Arg is within 4 Å, such Asn/Gln residue can be mutated to Asp/Glu to form a salt bridge. N372D, Q387E, and N402D were selected as this category. (2) Strengthening of existing salt bridges; in the case of salt bridges between Lys and Asp/Glu, such Lys residue can be mutated to Arg to enhance salt bridge because additional hydrogen bonds can construct between side chains. K273R was selected as this category. (3) Comparison with enzymes from other species; structure alignment between Ef-EG2 and *endo*-1,4- β -glucanase from termite *Nasutitermes takasagonesis* (NtEgl) (Khademi et al., 2012) shows that Asp43 in Ef-EG2 corresponds to Arg23 in NtEgl. Because Arg23 interacts electrostatically with Asp35 (Asp55 in Ef-EG2) in NtEgl, D43R can involve an electrostatic interaction, possibly with a salt bridge.

Site-directed mutagenesis

The K273R, N372D, Q387E, N402D, and D43R mutants were generated using the PrimeSTAR Mutagenesis Basal Kit (Takara Bio) based on the wild-type Ef-EG2 following the manufacturer's instructions. The oligonucleotides used as the forward and reverse primers to obtain the mutants are listed in Table S1. The plasmid pPICZ α -Ef-EG2 was used as a template. PCR reaction was carried out at 98 °C for 1 min, 98 °C for 10 s, 55 °C for 15 s, 72 °C for 10 s for 30 cycles, the reaction at 72 °C for 5 min, and storage at 4 °C. The PCR product was transformed into *E. coli* DH5 α competent cells and further confirmed by DNA sequencing.

Expression and purification of mutant enzymes in *pichia pastoris* GS115

The expression plasmid pPICZ α -Ef-EG2 mutants were linearized by *Pme*I and transformed into competent *P. pastoris* GS115 cells by electroporation. Transformed cells were spread on YPDS medium (1% yeast extract, 2% peptone, 2% dextrose, 1 M sorbitol, and 1.5% agar) containing 100 μ g/mL of Zeocin and incubated at 28 °C for 2–4 days. Colonies were spread onto YPDS plates containing 1000 or 2000 μ g/mL of Zeocin. Colonies capable of growth in the presence of a high concentration of Zeocin were selected. The selected colonies were then cultured in a 1 L Erlenmeyer flask containing 200 mL BMGY medium (1% yeast extract, 2% peptone, 100 mM potassium phosphate pH 6.0, 1.34% YNB, 4×10^{-5} % biotin, 1% glycerol) at 28 °C. The cultured media was centrifuged at 8,200 g for 10 min, and the resulting cell pellets were resuspended in BMMY medium (1% yeast extract, 2% peptone, 100 mM potassium phosphate pH 6.0, 1.34% YNB, 4×10^{-5} % biotin, 0.5% methanol). The cell suspension was then added to an Erlenmeyer flask containing 800 mL of BMMY medium and grown at 17 °C for 7 days at 200 rpm with daily additions of 0.5% methanol.

After cultivation, the cultured media was centrifuged at 15,400 $\times$ g for 30 min at 4 °C before the supernatant was recovered. Ef-EG2 mutants were loaded onto a HisTrapTM FF column (column volume: 1 mL) (GE Healthcare, Little Chalfont, Buckinghamshire, UK) equilibrated with binding buffer (20 mM Tris-HCl buffer (pH 8.0), 0.5 M NaCl, 20 mM imidazole). The enzyme was eluted with elution buffer (20 mM Tris-HCl buffer (pH 8.0), 0.5 M NaCl, 200 mM imidazole). The active fractions after HisTrapTM FF column chromatography were pooled and desalted on a PD-10 column (GE Healthcare, Little Chalfont, Buckinghamshire, UK). Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) was used to determine the purity of the recombinant proteins.

Enzyme and protein assay

The activity of Ef-EG2 was detected by the release of reduced sugar from CMC, as previously described Ueda et al. (2010). The amount of enzyme activity required to form an amount of reducing sugar corresponding to 1 μ mol of glucose per min of reaction time was one unit of enzyme activity. Protein concentration of Ef-EG2 protein was calculated using absorbance at 280 nm and the protein extinction coefficient, according to the method of Gilland von Hippel (Gill and von Hippel, 1989).

Effects of temperature on enzyme activity and enzyme stability

The reaction mixture consisted of 0.4% CMC in 0.1 M sodium acetate buffer (pH 5.0) and enzyme. After incubation for 15 min in the range of 10–60 °C, the amount of reducing sugar contained in the sample was then determined according to the Somogyi-Nelson method (Somogyi, 1952).

To measuring thermostability, each purified enzyme was incubated in 50 mM sodium acetate buffer (pH 6.0) for 30 min at various temperatures in the range of 10–60 °C. After incubation, the remaining activities were measured under standard assay conditions. The average values of triplicate measurements were used for each activity value. The stabilities of mutant enzymes were determined by measuring the residual activities after pre-incubation of each enzyme in 50 mM sodium acetate buffer (pH 6.0) at 50 °C for various periods.

Crystallization, diffraction data collection, and structure refinement

The solution containing each purified enzyme (D43R, N372D, and Q387E) was exchanged to 20 mM potassium phosphate pH 7.0. The samples were concentrated to the final concentration of 17 mg/ml (D43R), 15 mg/ml (N372D), and 10 mg/ml (Q387E) using an AmiconUltra 3-kDa molecular-weight cutoff filter (Merck). Initial screening was performed with Index (Hampton Research) and JCSG-plus (Molecular Dimensions) screening kits. The crystals used for X-ray diffraction data collection were prepared by the hanging-drop vapor-diffusion method at 20 °C using a reservoir solution consisting of 16–19%(w/v) PEG3350, 0.2 M magnesium chloride, and 50 mM sodium acetate pH 5.0.

X-ray diffraction experiments were performed on the BL5A beamline (D43R) and on the BL17A beamline (N372D and Q387E) of Photon Factory (KEK, Tsukuba, Japan). The crystals were soaked in a cryoprotectant solution containing the mother liquor supplemented with 20% (v/v) glycerol, and the crystals were flash cooled in liquid nitrogen. The wavelength was set to 1.0 Å (D43R) and 0.98 Å (N372D and Q387E). The diffraction data sets were collected at 100 K. The diffraction intensities were integrated using the program XDS (Kabsch, 2010) and scaled using the program Aimless (Evans and Murshudov, 2013).

Molecular replacement procedures were performed with the program Phaser (McCoy et al., 2007) using the structure of the wild type (PDB code: 3WC3; Arimori et al., 2013) as a search model. Structure refinement was performed using the program Phenix (Liebschner et al., 2019). Models and electron density maps were displayed and modified using the program Coot (Emsley et al., 2010). Figures were prepared

using the program Pymol (Schrödinger and DeLano, 2020).

Circular dichroism measurements

Temperature dependent changes in protein folding were monitored using J-720 CD spectrometer (JASCO). Each enzyme solution used for the measurement was prepared at the concentration of 0.14 mg/ml in 20 mM potassium phosphate pH 7.0. The wavelength was set to 220 nm, and the temperature step was set to 1 °C/min in the range of 10–95 °C. The measurements were performed using a 1 mm path length cuvette. The midpoint temperatures of the unfolding transition (T_m) were calculated as shown in the reference (Greenfield, 2006).

Molecular dynamics simulation

The structures of WT, D43R, N372D, and Q387E enzymes were prepared and processed using the “Structure Preparation” function in *Molecular Operating Environment* (MOE) (2022. 02, Chemical Computing Group ULC, Canada). The determination of protonation states was performed with the “Protonate3D” function in MOE (Labute, 2008). The enzyme molecules were placed in a periodic box of water molecules with a marginal distance of 10 Å, and sodium and chloride ions were added to neutralize the system. The Amber10:EHT force field was assigned to each system. Each system was equilibrated for 100 picoseconds at 300 K, and then simulations were carried out 100 ns at 300 K with a time step of

0.002 ps. The MD simulations were conducted using the PMEMD module of Amber18 (Case et al., 2018), and CPPTRAJ was used to perform the analysis of MD trajectories (Roe & Cheatham, 2013).

Results

Site-direct mutagenesis, expression, and purification of rEf-EG2 WT and mutant enzymes

rEf-EG2 WT and five mutant enzymes (K273R, N372D, Q387E, N402D, D43R) were purified using HisTrap FF column chromatography. The molecular masses of purified rEf-EG2 WT and mutant enzymes were estimated to be 51.7 kDa using SDS-PAGE (Fig. 1). The production of rEf-EG2 per liter (mg/L) was 1.95 (WT), 0.528 (K273R), 5.57 (N372D), 5.68 (Q387E), 5.06 (N402D), 5.66 (D43R), respectively. The two mutant enzymes (K273R, N402D) lost their cellulase activity.

Properties of rEf-EG2 WT and mutants

The functional properties of purified rEf-EG2 WT and mutant enzymes were determined by enzyme assays with CMC as the substrate. The optimal temperatures (40 °C) of the mutant enzymes were similar to that of the WT enzyme (Fig. 2-A). D43R enzyme had higher specific activities than N372D and Q387E enzymes at 50–60 °C. The mutant enzymes had the 2.5-fold higher specific activities than the WT enzyme

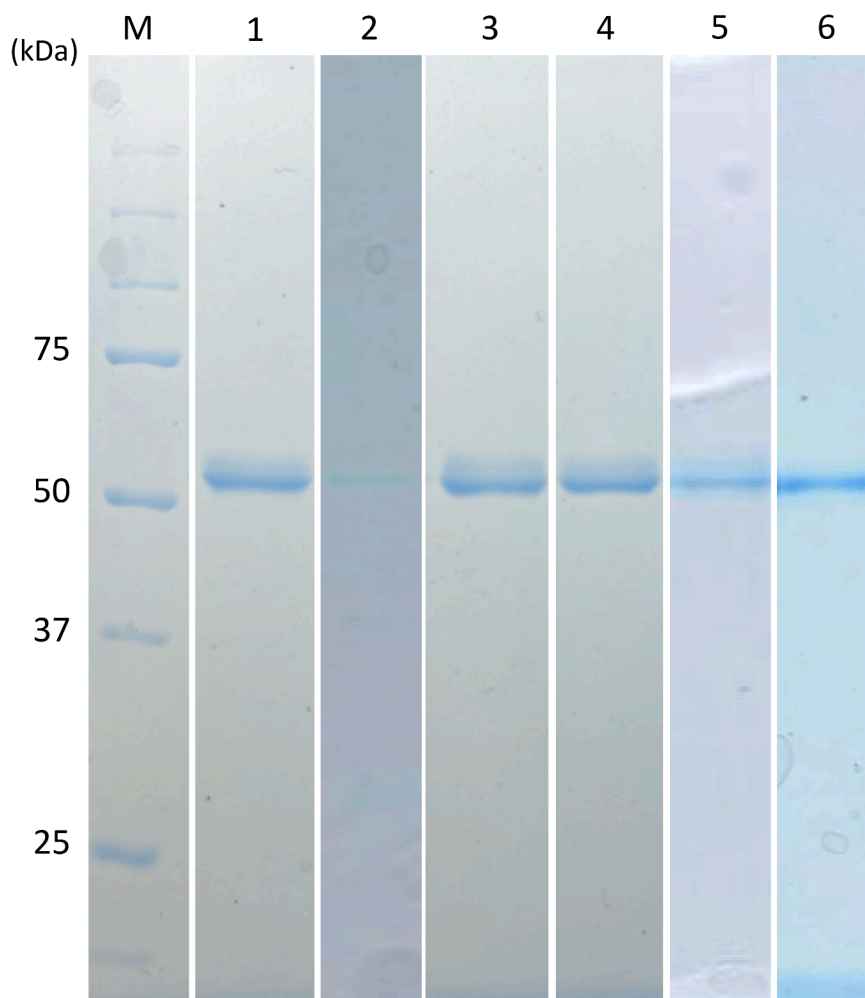


Fig. 1. SDS-PAGE analysis of Ef-EG2 WT and mutant enzymes. Lane M: Precision Plus Protein™ Unstained Standards (Bio-Rad Laboratories, Inc., CA, USA), Lane 1: WT, 2: K273R (1.4 µg protein), 3: N372D (1.0 µg protein), 4: Q387E (2.0 µg protein), 5: N402D (2.4 µg protein), 6: D43R (1.8 µg protein). Parentheses indicate the amount of sample applied.

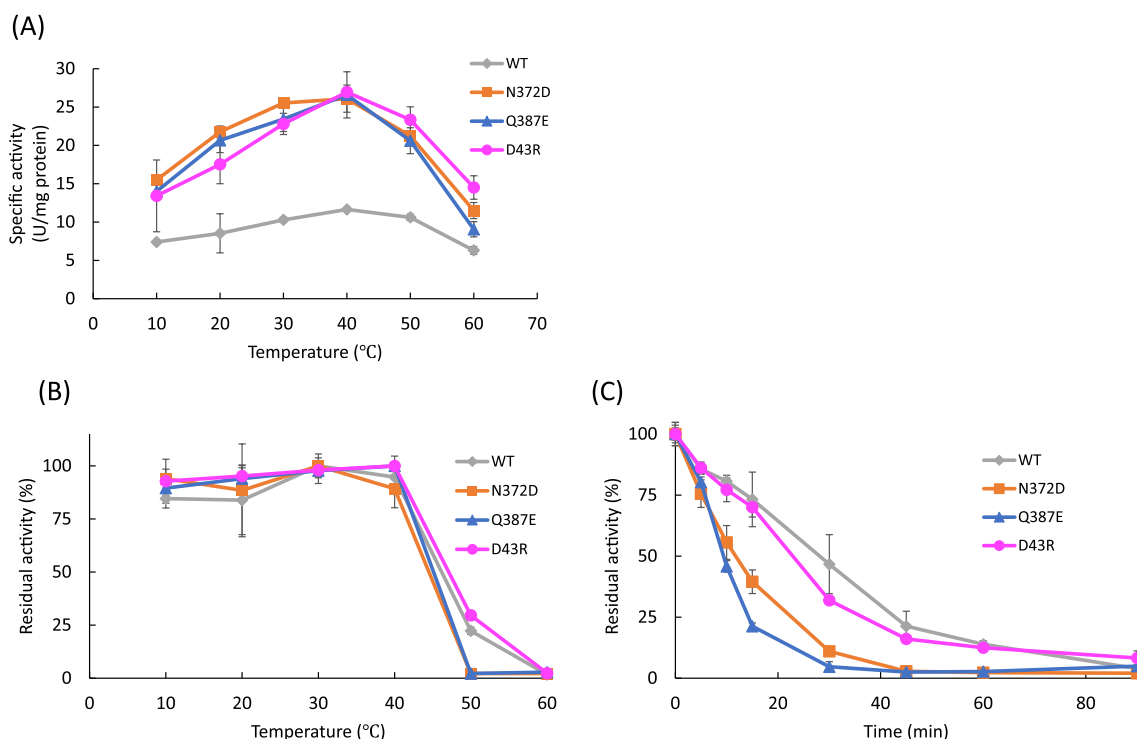


Fig. 2. Comparison of the enzymatic properties of the purified mutants N372D, Q387E, and D43R with the wild-type Ef-EG2. (A) Specific activities of the mutants and WT in various temperatures. (B) Effects of temperatures on stabilities of the mutants and WT. After the enzymes were pre-incubated at various temperatures, the residual activities were assayed at 37 °C in 50 mM acetate buffer (pH 6.0). (C) Thermostability of the mutants and WT at 50 °C. Residual activities were assayed at 37 °C after pre-incubation at 50 °C for different periods of time in 50 mM acetate buffer (pH 6.0).

over various temperatures (Fig. 2-A).

The thermal stabilities of the WT, N372D, Q387E and D43R enzymes showed 80% or more residual activity at 40 °C. The residual activities at 50 °C were 22.3%, 2.24%, 2.21%, and 29.6% for WT, N372D, Q387E and D43R (Fig. 2-B). It was found that the stabilities of the N372D and Q387E enzymes were significantly reduced. At 60 °C, all the enzymes were inactivated. The stability of the D43R enzyme showed the same tendency as that of the WT enzyme (Fig. 2-C). The residual activities of N372D and Q387E at 50 °C for 45 min were also significantly reduced.

CD spectroscopy for the analysis of temperature dependent unfolding

The ellipticities (θ) at the wavelength of 220 nm against temperature are plotted for WT, D43R, N372D, and Q387E (Fig. 3). The crystal structure of Ef-EG2 WT (PDB code: 3WC3) shows an enrichment of α -helices (54% of the whole sequence). The ellipticity at 220 nm of an α -helical protein is lower than that of a denatured protein. Therefore, increases in the ellipticities as a function of temperature indicate a transition from folding to unfolding of proteins. The midpoint temperatures of the unfolding transition (T_m) are 55.1 °C (WT), 55.6 °C (D43R), 51.5 °C (N372D), and 51.2 °C (Q387E). This result shows that

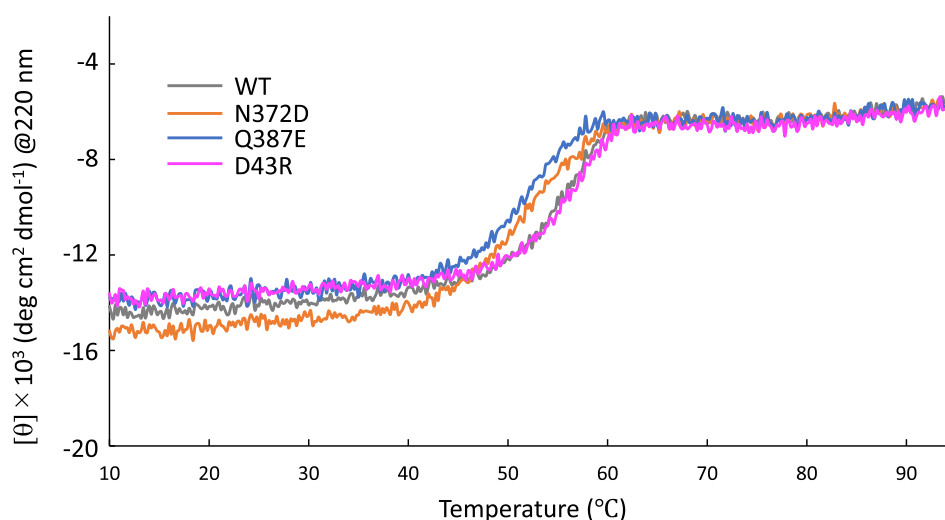


Fig. 3. Temperature dependent unfolding transition of Ef-EG2. The ellipticities (θ) at the wavelength of 220 nm in CD measurements are plotted against temperature range of 10–95 °C for WT (gray), D43R (magenta), N372D (orange), and Q387E (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

thermostabilities of N372D and Q387E are lower than that of WT, while D43R is the same thermostability level of WT.

Crystal structure of mutant enzymes

To investigate structural features affecting the catalytic activities and thermostabilities, X-ray crystal structures were determined at resolutions of 1.7 Å (D43R), 1.6 Å (N372D), and 1.6 Å (Q387E) (Table S2). The superposition of the overall structures among WT, D43R, N372D, and Q387E showed high similarities (Fig. 4-A, Table S3). The structures of the catalytic site also showed high similarities (Fig. 4-B). These mutation sites were selected at the protein surface on the opposite side of the substrate binding cleft (Fig. 4-C).

At the mutation site of Asp43, the WT structure shows that a sodium ion is coordinated by the side-chain carboxylates of Asp43 and Asp55, the main-chain carbonyl oxygen of Leu44, and three water molecules (Fig. 5-A). However, a corresponding sodium ion coordination is not observed by the arginine mutation of Asp43. The D43R structure shows that the side-chain guanidino group of Arg43 is located at 3.82 Å from the carboxylate group of Asp55 (Fig. 5-B). The side-chain conformations of Arg43 and Asp55 indicate that the two residues do not form a hydrogen bond, but they are located within an electrostatic interaction distance.

At the mutation site of Asn372, the WT structure shows that the main-chain carbonyl oxygen of Asn372 forms hydrogen bonds with the

side-chain guanidino group of Arg363 (Fig. 5-C). In the N372D structure, the mutation to aspartic acid does not introduce a hydrogen bond between the side-chain carboxylate group of Asp372 and the side-chain guanidino group of Arg363 (Fig. 5-D). The carboxylate group of Asp372 is located within an electrostatic interaction distance from one conformation of the guanidino group of Arg363.

At the mutation site of Gln387, the WT structure shows that the side-chain of Gln387 does not form a hydrogen bond with the side-chain of Arg149, whereas the side-chain carboxylate group of Asp152 forms a salt bridge with the guanidino group of Arg149 (Fig. 5-E). In the Q387E structure, the side-chain carboxylate group of Glu387 does not form a hydrogen bond with the guanidino group of Arg149, but they are located within an electrostatic interaction distance (Fig. 5-F). The salt bridge between Asp152 and Arg149 is still maintained in the Q387E structure.

MD simulations of rEf-EG2 enzymes

The 100 ns MD simulations were performed for WT, D43R, N372D, and Q387E enzymes to analyze effects of the remote mutations on the catalytic residues. The distances of Asp74 Oδ2 – Asp77 Oδ2, Asp74 Oδ2 – Glu431 Oε2, and Asp77 Oδ2 – Glu431 Oε2 were selected to examine the dynamics of the catalytic residues (Fig. S1-A). The fluctuation of the distances did not show significant differences in the WT, D43R, N372D, Q387E enzymes during the simulations (Figs. S1-B, C and D).

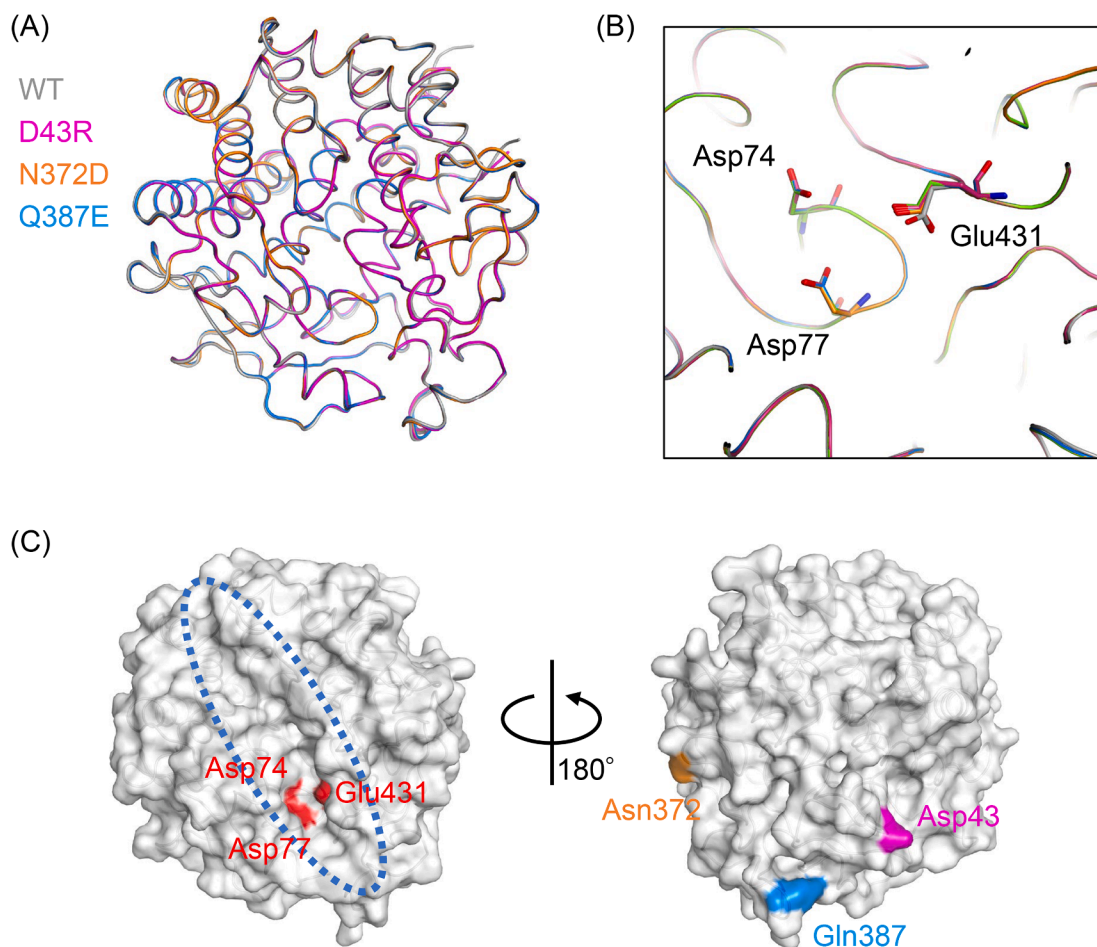


Fig. 4. X-ray crystal structure of Ef-EG2. (A) Superposition of overall structures among WT (gray), and D43R (magenta), N372D (orange), and Q387E (blue). The structures are shown as ribbon models. (B) Superposition of the structures around the catalytic site. Putative catalytic residues (Asp74, Asp77 and Glu431) are shown as stick models. (C) Protein surface of WT. The right panel represent the 180° rotation around the vertical axis of the left panel. Putative catalytic residues are colored as red, and the substrate binding cleft is indicated as a blue dotted circle in the left panel. The positions of Asp43 (magenta), Asn372 (orange), and Gln387 (blue) are shown in the right panel. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

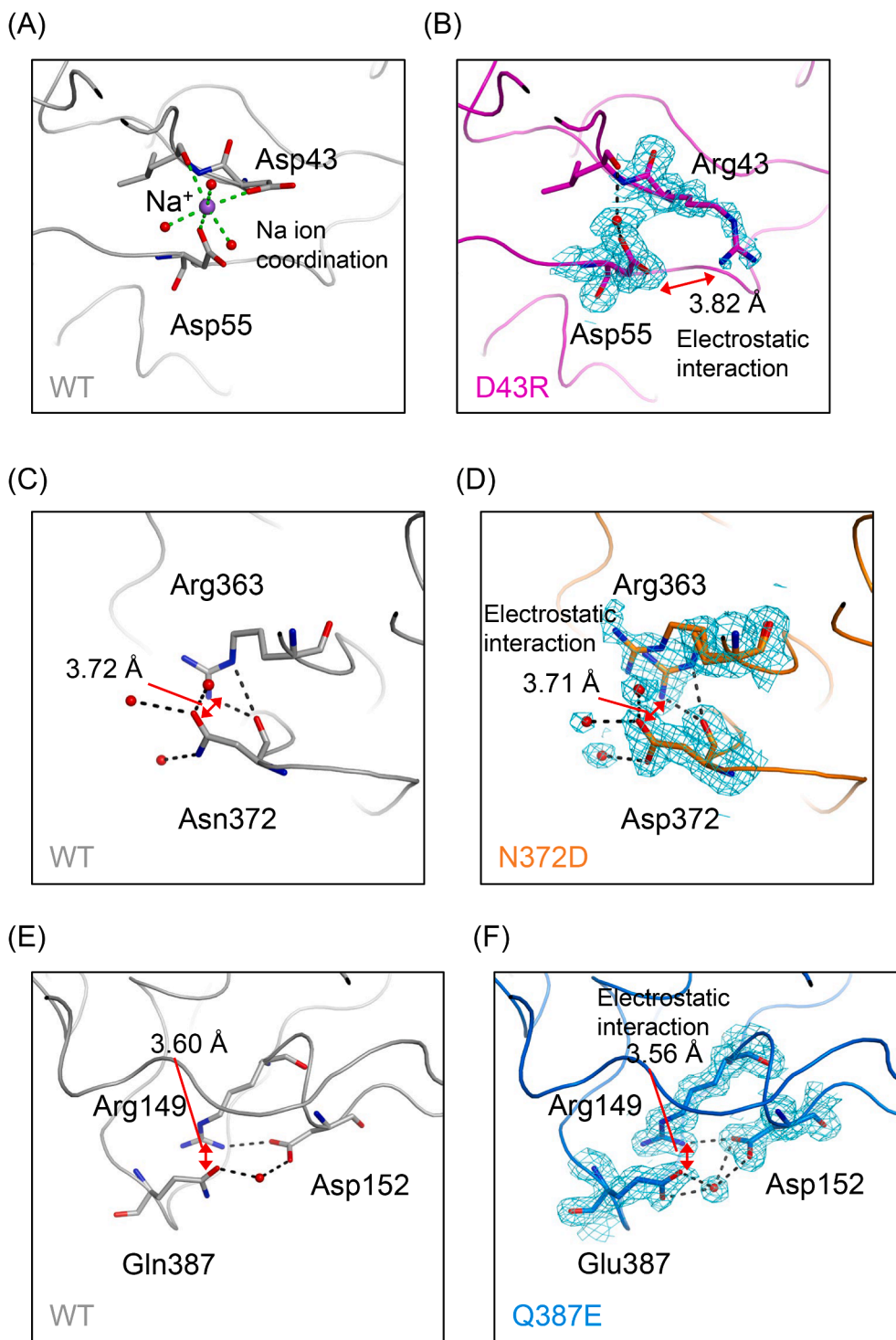


Fig. 5. The structures around the mutation sites. (A) The structure around Asp43 in WT. Asp43, Leu44, and Asp55 are shown as stick models. A sodium ion and water molecules are shown as purple sphere and red sphere, respectively. The sodium ion coordination is indicated by green dotted lines. (B) The structure around Arg43 in D43R. The distance between N^{ε1} of Arg43 and O^{δ1} of Asp55 are indicated in the figure. (C) The structure around Asn372 in WT. Asn372 and Arg363 are shown as stick models. (D) The structure around Asp372 in N372D. The distance between N^{ε2} of Arg363 and O^{δ2} of Asp372 are indicated in the figure. (E) The structure around Gln387 in WT. Gln387, Arg149, and Asp152 are shown as stick models. (F) The structure around Glu387 in Q387E. The distance between N^{ε2} of Arg149 and O^{δ2} of Glu387 are indicated in the figure. Hydrogen bonds are indicated as black dotted lines in figures B-F. 2Fo – Fc electron density map contoured at the 1.0 σ level is shown as a blue mesh in figures B, D, and F. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Discussion

The two mutant enzymes (K273R, N402D) lost cellulase activity. Lys273 is located near the Ca²⁺-binding amino acid (Glu234), and affects the Ca²⁺-binding region. It is suggested that the K273R enzyme lost the ability to combine Ca²⁺ by the mutation. Ca²⁺ is revealed stabilizing factor in amylase from *Bacillus stearothermophilus* (Pan et al., 2019). Identically, Ca²⁺ greatly affects on the stability of Ef-EG2 (Arimori et al., 2013). It is considered that the N402D enzyme influenced on the structure of active site and lost its activity because N402 was located in 10 Å from the active site (Glu431).

The remaining mutation sites (Asp43, Asn372, and Gln387) are distant from the catalytic site (Fig. 4-C), and the structures of the catalytic site do not show significant differences among WT, D43R, N372D, and Q387E (Fig. 4-B). D43R, N372D, and Q387E enzymes were improved the specific activities compared with WT enzyme (Fig. 2-A). One possibility for the improvement is that the mutations influence the binding of the substrate and/or release of the product during the catalytic reaction (Okuda et al., 2020). Another possibility is that the mutations influence the conformations of the active site residues (Jimenez-Oses et al., 2014). The MD simulations of WT, D43R, N372D, and Q387E enzymes did not show significant differences in the conformational

dynamics of the catalytic residues (Fig. S1). Nanosecond simulations are probably not long enough to analyze the effects of the remote mutations (Jimenez-Oses et al., 2014). Additional kinetic analyses and long-timescale MD simulations may explain the improvement in the specific activities.

Results of residual activities (Fig. 2-B, C) and CD spectroscopy (Fig. 3) indicate that WT and D43R possess similar thermostability level. The optimization of electrostatic interactions on the protein surface has been reported to improve the protein thermostability (Perl & Schmid, 2001). The arginine mutation introduces the additional electrostatic interaction with Asp55, and thus the surface charge optimization may contribute to the protein thermostability. The sodium ion coordination in the WT structure probably offers the improvement of thermostability because a sodium ion has been suggested to function as a stabilizing factor for protein structure (Ban et al., 2017). D43R enzyme has higher specific activities than N372D and Q387E enzymes at 50–60 °C (Fig. 2-A). These results may be attributed to differences in thermostabilities (Fig. 2-B, C and 3).

N372D and Q387E enzymes show lower thermostability compared with WT enzyme (Fig. 2-B, C and 3). Mutational and computational studies of protein thermostability have shown that decrease of negative surface charge increases the protein thermostability (Torrez et al., 2003). Therefore, the negative surface charge introduced by the mutation of N372D or Q387E may decrease the thermostability of Ef-EG2. Indeed, N372D and Q387E enzymes show lower thermostability level compared with WT and D43R enzymes (Fig. 2-B, C and 3). N372D and Q387E enzymes have higher specific activities than D43R enzyme at 10–20 °C (Fig. 2-A). Enzymes from *E. fetida*, which possess catalytic activities at low temperatures, have been reported to possess larger fractions of surface area with negative potentials compared with their thermophilic counterpart (Arimori et al., 2013, Hirano et al., 2020). Introducing a negative charge on the protein surface can further enhance the activity at low temperatures.

Conclusion

In this study, we obtained the mutant enzymes (N372D, Q387E, and D43R) that have 2.5-fold higher specific activities than WT. All the mutation sites are distant from the catalytic site, and the structures of the catalytic site do not show significant differences among WT, D43R, N372D, and Q387E. It is possible that the mutations of residues distant from the active site changed the efficiency of the interaction with substrate, and improved the catalytic activity. The mutant enzymes seem to be changed the conformational dynamics of the catalytic residues. Among the mutants, D43R showed almost the same thermostability as the WT enzyme. D43R enzyme lost the ability to bind sodium ions, but was introduced the additional electrostatic interaction with Asp55. N372D and Q387E enzymes show higher specific activities at low temperature region and lower protein thermostabilities than D43R enzymes, and thus the surface charge optimization may contribute to the protein thermostability and specific activity.

CRediT authorship contribution statement

Chikako Kuroki: Writing – original draft, Data curation, Investigation. **Yu Hirano:** Writing – original draft, Writing – review & editing, Data curation, Investigation. **Mami Nakazawa:** Supervision. **Tatsuji Sakamoto:** Supervision. **Taro Tamada:** Supervision, Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Mitsuhiko Ueda:** Supervision, Conceptualization, Data curation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.crbiot.2023.100126>.

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